# DA13 NumPy: the numbers underneath pandas

Week 5, Explore with NumPy and seaborn | Session 13 of 24 | Python for Data Analysis | New library: NumPy

**Today's question.** How spread out are the daily customer counts, and which sales are unusually large?

**Objective.** By the end: you can turn a column into a NumPy array, calculate its mean, median, spread and percentiles, and find unusually large values.

> **WHERE THIS IS USED**
> Weather offices and football analysts work with long rows of numbers, such as daily rainfall or distance run. They check how spread out the numbers are and spot unusual days. Which day in a shop would you call unusual?

Before this session: DA12 Show your first chart and explain what it says. If you cannot explain the last example, use its error card with the instructor first.

## Words for today

| Word | What it means | Think of it like this |
|---|---|---|
| NumPy | The library of fast number tools that pandas is built on. We load it as np. | The calculator inside the register. |
| array | A row of numbers that NumPy can calculate with all at once. | A strip of numbers on a ruler. |
| element-wise | Doing the same calculation to every number in an array at once. | Adding 100 shillings to every price on the board. |
| percentile | The value below which a given share of the numbers fall. | The mark that 75 out of 100 pupils did not beat. |
| standard deviation | A number for how far values typically sit from their mean. | How far pupils usually stand from the middle of the line. |
| outlier | A value far from the rest. | One huge order in a week of small sales. |

## Block A. Arrays, summaries and percentiles

Each shop counted its customers every day for 60 days: 180 counts in all. .to_numpy() turns a column into an array. np.mean and np.median give the usual summaries and np.percentile gives the value below which a share of the counts fall.

**Predict.** Predict whether the mean or the median of the daily counts is bigger. Write your guess on paper first.

*Your guess, written before you run it*

```
import numpy as np
import pandas as pd
daily = pd.read_csv("sunrise_daily_summary.csv")
customers = daily["customers_counted"].to_numpy()
print(customers.shape)
print(round(np.mean(customers), 1), np.median(customers))
print(np.percentile(customers, [25, 75]))
print(customers.max() - customers.min())
print(np.mean(customers) == daily["customers_counted"].mean())
```

**Run it, then compare.** The array has shape (180,). The mean is 143.0 and the median 131.5, so the mean is bigger. The 25th and 75th percentiles are 86.75 and 203.75: a quarter of the days had fewer than 87 customers and a quarter had more than 204. The range is 332. The last line prints True, because pandas and NumPy give the same mean.

**Investigate.**

1. Why are the mean and the median different here?
2. What do the two percentiles tell the owner?

**Modify.** Add 50 to every count at once and print the new mean.

## Block B. Spread and outliers

The standard deviation says how far values usually sit from their mean. A common rule marks a value as an outlier when it is more than 1.5 times the gap between the quartiles above the upper quartile.

**Predict.** Predict how many of the 1513 sales are unusually large by that rule. Write your guess on paper first.

*Your guess, written before you run it*

```
sunrise = pd.read_csv("sunrise_sales.csv")
values = (sunrise["units"] * sunrise["unit_price_ugx"]).to_numpy()
q1, q3 = np.percentile(values, [25, 75])
fence = q3 + 1.5 * (q3 - q1)
big = values[values > fence]
print(q1, q3, fence)
print(len(big), big.max())
print(round(np.std(customers), 1))
```

**Run it, then compare.** The quartiles are 1200 and 4500, so the fence is 9450. 137 of the 1513 sales are above it, and the largest is 108000. The standard deviation of the daily customer counts is 73.7. A few large sales sit far out. They are real sales, so we report them and do not delete them.

**Investigate.**

1. Should we delete the 137 outliers?
2. What does a standard deviation of 73.7 customers tell us?

**Modify.** Count the sales with a value above 20000.

## Make

Take the revenue_ugx column of the daily summary as a NumPy array. Find its mean, median and the 25th and 75th percentiles with NumPy. Then find the same four numbers with pandas and check that the two routes agree.

Save your work as DA13. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

## Error card

| You see | It means | Fix |
|---|---|---|
| TypeError: 'tuple' object is not callable | shape is a value you read. It takes no brackets, and brackets try to call it like a function. | Remove the brackets after shape. |
| The mean from NumPy and pandas differ in the last digits | Decimals are stored with tiny rounding differences. | Compare with np.allclose, or round both to the same number of places. |

## Exit ticket

NumPy gives me the numbers underneath a column, and an outlier is a real value I report, not one I delete.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.